

Scalable and sequential inference of the neutron star equation
of state with the Einstein Telescope

Thibaud Wouters



Utrecht
University



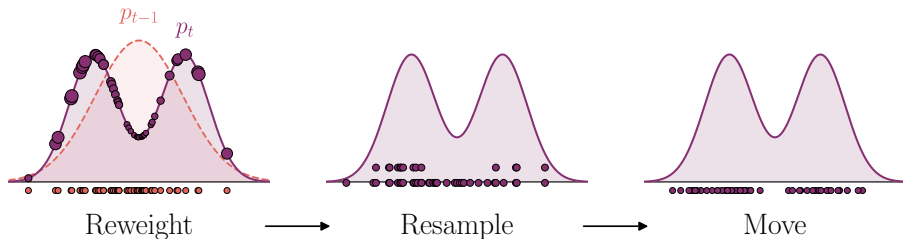
Equation of state inference in the 3G era

- Tidal deformations in BNS mergers probe the neutron-star equation of state (EOS)
- Einstein Telescope/Cosmic Explorer: $\mathcal{O}(10^4)$ BNS mergers per year [1]
- **Bottleneck**: existing methods must restart from scratch as the catalog grows, we need
 - Scalable: e.g. GPU-acceleration JESTER
 - **Sequential**: reuse previous results, avoid restarting from the prior

SMC

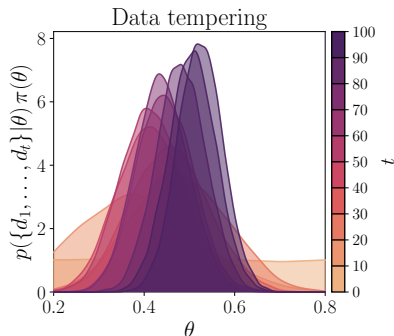
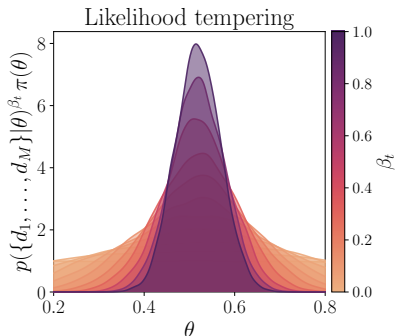
Sequential Monte Carlo (SMC)

- Particles evolve prior $\rightarrow$ posterior through a sequence of intermediate densities, via **reweight**, **resample**, **MCMC move**



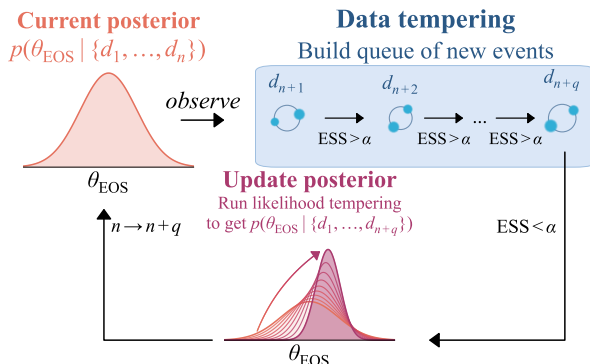
Likelihood vs. data tempering

- **Likelihood tempering**: anneal in the full likelihood, $\mathcal{L}(\theta)^{\beta_t}$, $\beta_t : 0 \rightarrow 1$
- **Data tempering**: anneal in new observations one at a time



Adaptive hybrid algorithm

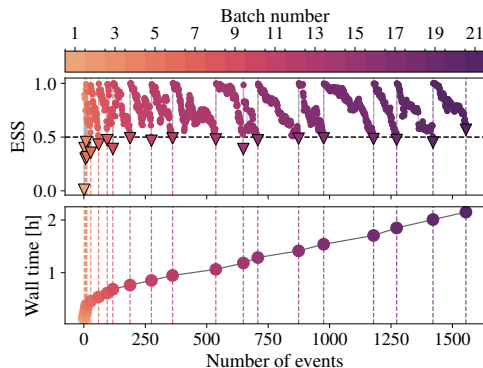
- New events **queued** while effective sample size (ESS) stays high
- ESS drops below threshold $\alpha \Rightarrow$ **update** posterior with likelihood tempering over the batch
- Naturally handles **growing catalogs** without restarting from the prior



Simulating one month of ET data

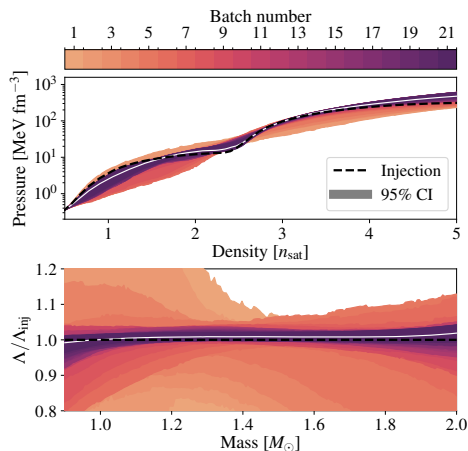
- ET 2L-misaligned configuration, Gaussian noise, merger rate $106.6 \text{ Gpc}^{-3} \text{ yr}^{-1}$
- $\text{SNR} > 12 \Rightarrow 1555$ detected BNS sources
- Single-event PE: nested sampling (DYNESTY [2]), masses + tidal deformabilities only, $f_{\text{low}} = 20 \text{ Hz}$
- SMC settings: $N = 2000$ particles, ESS threshold $\alpha = 50\%$, prior informed by PSR J0740+6620 [3]

SMC diagnostics



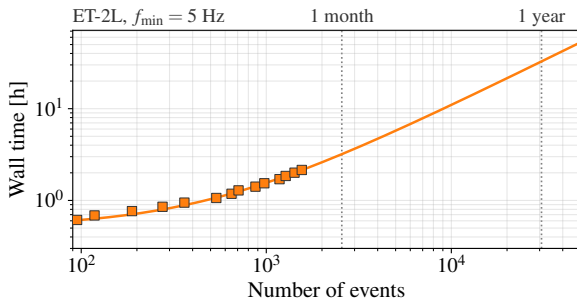
- ESS **below 50%** triggers an update (∇)
- Early events highly informative: **frequent** updates
- Later batches: hundreds of events before the next update
- Total: **22** updates, ~ 2.2 h on 1 NVIDIA H100 GPU

Equation of state constraints



- Bulk region ($1-3 n_{\text{sat}}$):
pressure
under-/over-estimated
- Injected Λ within the 95%
credible band throughout
- Final batch: $\sim 1\%$
constraints on Λ across
probed masses

Extrapolating to a full year



- Runtime flat up to $N \sim 200$, then memory-bound $\Rightarrow$ roughly linear fit ($N > 400$)
- More realistic settings ($f_{\min} = 5$ Hz): ~ 2559 events/month, $\sim 30\,000$ /year
- Projected runtime: **a few hours** (1 month), **~ 1.5 days** (1 year) on a single GPU

Conclusion & outlook

- Recipe for scalable, sequential hierarchical inference: GPU + normalizing flows + SMC
- Demonstrated on a realistic $\mathcal{O}(10^3)$ -event ET mock catalog
- Extrapolation: a full year ($\sim 30\,000$ events) feasible within days on a single GPU
- Outlook: joint EOS + population + cosmology inference [4]; flow surrogate for compounded hyperlikelihood cost

Thanks for listening!



References I

- [1] Michele Maggiore et al. “Science Case for the Einstein Telescope”. In: *JCAP* 03 (2020), p. 050. DOI: [10.1088/1475-7516/2020/03/050](https://doi.org/10.1088/1475-7516/2020/03/050). arXiv: [1912.02622](https://arxiv.org/abs/1912.02622) [astro-ph.CO].
- [2] Joshua S. Speagle. “dynesty: a dynamic nested sampling package for estimating Bayesian posteriors and evidences”. In: *Mon. Not. Roy. Astron. Soc.* 493.3 (2020), pp. 3132–3158. DOI: [10.1093/mnras/staa278](https://doi.org/10.1093/mnras/staa278). arXiv: [1904.02180](https://arxiv.org/abs/1904.02180) [astro-ph.IM].
- [3] E. Fonseca et al. “Refined Mass and Geometric Measurements of the High-mass PSR J0740+6620”. In: *Astrophys. J. Lett.* 915.1 (2021), p. L12. DOI: [10.3847/2041-8213/ac03b8](https://doi.org/10.3847/2041-8213/ac03b8). arXiv: [2104.00880](https://arxiv.org/abs/2104.00880) [astro-ph.HE].
- [4] Hauke Koehn et al. “Binary neutron stars in the next-generation era: Multi-messenger detection prospects and constraints on the equation of state, mass distribution, and cosmology”. In: (July 2026). arXiv: [2607.28438](https://arxiv.org/abs/2607.28438) [astro-ph.HE].
- [5] Peter Neal and Gareth Roberts. “Optimal scaling of random walk Metropolis algorithms with non-Gaussian proposals”. In: *Methodology and Computing in Applied Probability* 13.3 (2011), pp. 583–601.

Backup: SMC — likelihood tempering

$$p_t(\theta) = \frac{\mathcal{L}(\theta)^{\beta_t} \pi(\theta)}{\mathcal{Z}_t}, \quad w_t^i = \mathcal{L}(\theta_{t-1}^i)^{\beta_t - \beta_{t-1}}$$

$$\beta_t = \sup_{\beta \in [\beta_{t-1}, 1]} \{\text{ESS}_t(\beta) \geq \alpha N\}$$

```
1:  $t \leftarrow 1$ ,  $\beta_1 \leftarrow 0$ ,  $\hat{\mathcal{Z}}_1 \leftarrow 1$ 
2: for  $i = 1, \dots, N$  do
3:    $\theta_1^i \sim \pi(\theta)$  ▷ sample from the prior
4: end for
5: while  $\beta_t < 1$  do
6:    $t \leftarrow t + 1$ 
7:    $\beta_t \leftarrow \sup_{\beta \in [\beta_{t-1}, 1]} \{\text{ESS}_t(\beta) \geq \alpha N\}$  ▷ adaptive temperature
8:   for  $i = 1, \dots, N$  do
9:      $w_t^i \leftarrow \mathcal{L}(\theta_{t-1}^i)^{\beta_t - \beta_{t-1}}$  ▷ reweight
10:  end for
11:   $\hat{\mathcal{Z}}_t \leftarrow \hat{\mathcal{Z}}_{t-1} \times \frac{1}{N} \sum_i w_t^i$  ▷ update evidence
12:  resample  $\{\theta_{t-1}^i\}$  with weights  $\{w_t^i\}$ 
13:  move particles with MCMC kernel  $\mathcal{K}_t$  ▷ rejuvenation
14: end while
```

Settings: $N = 2000$, $\alpha = 0.9$, Gaussian random-walk kernel, 20 MCMC steps, target acceptance 0.234 [5].

Backup: SMC — data tempering

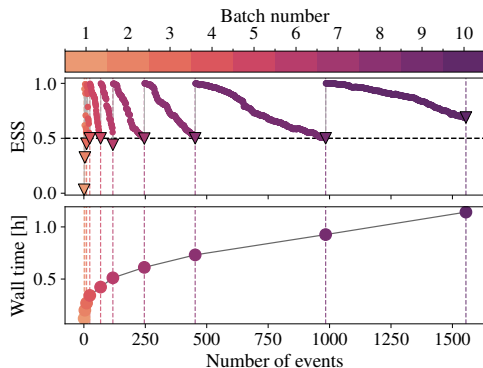
$$\mathcal{P}_t(\theta) = \frac{\mathcal{L}(d_{1:b_t}|\theta)\pi(\theta)}{\mathcal{Z}_t}, \quad w_t^i = \mathcal{L}_{b_{t-1}:b_t}(\theta_{t-1}^i)$$

$$b_t = \sup_{b \in [b_{t-1}, T]} \{\text{ESS}_t(b) \geq \tilde{\alpha}N\}$$

```
1:  $t \leftarrow 0, n \leftarrow 0, q \leftarrow 0, \hat{\mathcal{Z}}_0 \leftarrow 1$ 
2: for  $i = 1, \dots, N$  do
3:    $\theta_0^i \sim \pi(\theta), w_0^i \leftarrow 1/N$ 
4: end for
5: while  $t < M$  do
6:    $t \leftarrow t + 1, q \leftarrow q + 1$ 
7:   for  $i = 1, \dots, N$  do
8:      $w_t^i \leftarrow w_{t-1}^i \times \mathcal{L}_t(\theta_n^i)$  ▷ reweight with new event
9:   end for
10:  if  $\text{ESS}_t < \tilde{\alpha}N$  or  $t = M$  then
11:     $\{\theta_t^i\}, \hat{\mathcal{Z}}_{n:n+q} \leftarrow \text{Alg. 1}(\mathcal{P}_n, \mathcal{L}_{n:n+q})$  ▷ batch update
12:     $\hat{\mathcal{Z}}_t \leftarrow \hat{\mathcal{Z}}_n \times \hat{\mathcal{Z}}_{n:n+q}, n \leftarrow n + q, q \leftarrow 0$ 
13:     $w_t^i \leftarrow 1/N$  ▷ clear the queue
14:  end if
15: end while
```

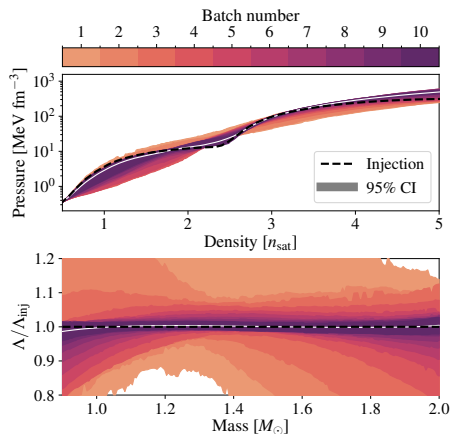
Threshold $\tilde{\alpha} = 0.5$; inner likelihood-tempering step uses the same settings as Algorithm 1.

Backup: zero-noise SMC diagnostics



- Same catalog, **no detector noise**
- ESS decays **smoothly and monotonically**, no erratic drops
- Fewer update steps needed: **10** batches vs. 22 with noise
- Lower total wall time as a result

Backup: zero-noise EOS constraints



- Same catalog, **no detector noise**
- Median Λ curve matches the injection **much more closely**
- Confirms that detector noise, not catalog size, drives the small bias seen in the noisy run